ComputerSD: Online Self-Distillation from Real-Time Feedback for Computer-Use Agents

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Paper version: arXiv 2609.40253v2 · CUA Radar Reel poster, not made or endorsed by the authors

Problem

> *Sparse outcome rewards provide no supervision for intermediate actions.*

Self-distillation adds token-level signal, but fixed guidance drifts from the agent's state, and the shifts it induces can contradict whether the step was correct.

Motivation: three open gaps

1. **Sparse credit.** A verdict does not rank the steps.
2. **Fixed guidance.** Hints drift from the state reached.
3. **Noisy shifts.** Over 80% of correct-step tokens are suppressed.

Evaluation setup

OSWorld-Verified · 361 | WindowsAgentArena · 154 | Qwen3-VL-8B-Thinking | EvoCUA-8B | Kimi K3 expert | 30-step train | 50-step eval | 3 evaluation runs

- **All online**; no offline step accuracy.
- **Held out:** Chrome + Multiple Apps (139 tasks).

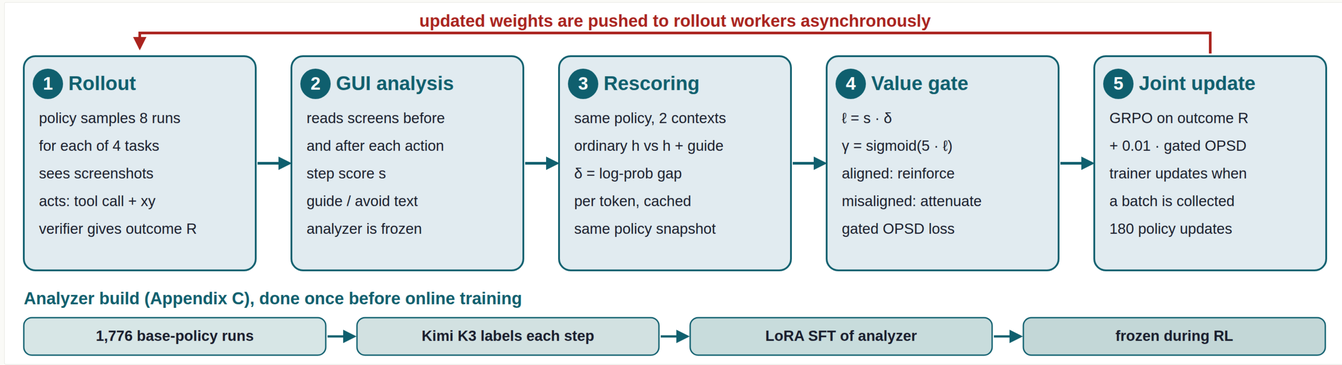
Our read (CUA Radar)

69	★★	transfer
score of 100	band	evidence level

- **Reports:** online eval; held-out categories; same-budget outcome-only GRPO; cost; component ablation; Windows transfer.
- **Not reported:** variance (no std/CI); analyzer check against ground truth. **Flag:** AP4, unclear.
- Pattern K19, evidence level transfer. Authors' numbers, not rerun.

Method: step feedback gates self-distillation, beside GRPO

- **Sees / acts:** screenshots (last 3) in; a computer_use tool call with coordinates out.
- **Signal:** frozen GUI analyzer scores each step and writes guidance.
- **Gate:** each token shift is weighted by agreement with the step score.

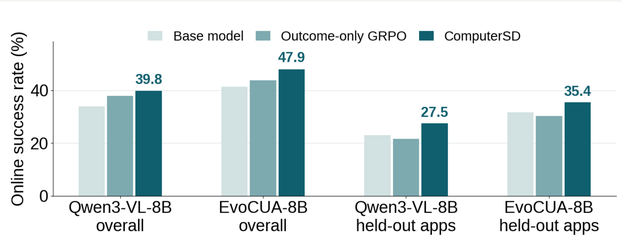


Our redraw (not the paper's figure): the online loop and the one-off analyzer build from the appendix.

$$\mathcal{L} = \mathcal{L}_{\text{GRPO}} + \lambda_{\text{OPSD}}\,\mathbb{E}[\gamma\,\ell], \quad \ell = s\,\delta,\ \gamma = \sigma(\beta_{\text{gate}}\,\ell)$$

joint loss; λ = 0.01, β_gate = 5; δ = log-prob gap

Key results: OSWorld-Verified online success (%)



Tables 1-2, redrawn: mean of 3 evaluation runs, 50 steps, no std.

+1.9 / +4.1 pts over GRPO overall (Qwen3 / EvoCUA)

SO WHAT → *Held-out apps = Chrome and Multiple Apps categories, not used in training. Other models' rows are in the paper's Table 1.*

Ablation: Qwen3-VL-8B OSWorld-Verified (%)



Table 3 and Table 11, redrawn: one component changed at a time.

SO WHAT → *Without real-time feedback (−4.8) or value gate (−3.2), success falls below outcome-only GRPO (37.9). Other gate designs in Table 4: SDAR 38.1, hard 35.2, reverse 36.0. Prompt-only guidance (no policy update) reaches 34.6.*

Headline Numbers

47.9%
EvoCUA-8B, OSWorld-Verified
vs 43.8% for outcome-only GRPO

39.8%	Qwen3-VL-8B overall
35.4%	EvoCUA held-out apps
+1.9 / +4.1	pts vs GRPO (paper)

Controls reported

CONTROL (ONLINE UNLESS NOTED)	COMPUTERSD	COMPARED WITH
In-domain: Qwen3 / EvoCUA	47.5 / 55.7	GRPO 48.2 / 52.2
WindowsAgentArena (Q / E)	24.5 / 27.8	GRPO 23.3 / 24.2
Analyzer vs expert, offline	85.3% agree	base model 29.9%
Trajectories per hour	about 210	GRPO about 370

Takeaway

Analyzer feedback gates self-distillation beside GRPO: +1.9 / +4.1 points, larger on held-out apps; Qwen3 in-domain 47.5 vs GRPO 48.2.

3 runs, no std | Kimi K3 labels | 56% / 22% of GRPO trajectories

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Paper | Code